

# Javanese numerals in composition\*

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**Abstract** Numerals have been proposed to live two semantic lives: they can denote primitive numbers or properties of individuals. Which denotation is basic and which is derived remains an open question. In Javanese (Austronesian, Malayo-Polynesian), morphologically simple numerals flag the number-property duality in morphosyntax: number-denoting numerals immediately precede the unit noun of which they are an argument, and they bear a linker suffix; property-denoting numerals linearly follow the noun being counted over, and they do not bear the linker. I sharpen existing descriptions and show that standard tools of semantic composition readily account for morphologically simple numerals. The core puzzle of this paper concerns complex numerals: in Javanese, numerals like ‘thirty’ have the morphosyntax of measurement phrases (MPs) like ‘three kilograms’ but have the semantic workload of simple numerals like ‘three.’ Between the analytical options of a basic-number and basic-property denotation for numerals, I argue against a unilateral treatment of all Javanese numerals having basic-number denotations. I offer a family of solutions that account for the morphosyntactic asymmetries between simple numerals and complex numerals. In its broader implications, the Javanese data pose a challenge for numeral analyses that universally endow numerals with basic number denotations. Along the way, I also argue: (i) simple numerals’ morphological alternations map to a measurement-individuation distinction; (ii) complex numerals, as with all MPs in the language, are constituents; and (iii) the numeral linker suffix is sensitive to hierarchical, not linear, structure.

## 1. Introduction

### 1.1. The double life of numerals

This paper analyzes the relationship between cardinal numerals’ morphosyntax and their interpretation in Javanese, specifically in a nonstandard variety spoken in Pemalang, Central Java. For at least fifty years, an adjectival theory of numerals has been on the market.<sup>1</sup> In general, this theory

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<sup>1</sup>See [Buccola 2015](#), Ch. 1, fn. 10 for a summary of early works on adjectival theories of numerals.

considers numerals to behave like adjectives rather than like determiners. A particularly influential adjectival treatment of numerals comes from Landman (2004). Among the empirical challenges for an earlier numeral-as-determiner analysis of English numerals are that numerals can co-occur with determiners like *the* and that they can follow adjectives:

- (1) a. We shipped the *fifty ferocious lions* to Blijdorp, and the *thirty meek lions* to Artis.  
 b. We shopped the *ferocious fifty lions* to Blijdorp, and the *meek thirty lions* to Artis.  
 (Landman 2004, 3)

In light of (1), analyzing *the fifty* as a complex determiner in (1a) invites the possibility that *the ferocious* is a complex determiner in (1b). Instead of complicating the possibilities for potential determiners, Landman (2004) proposes that numerals in (1) denote properties, as in (2):

- (2)  $\llbracket \text{fifty} \rrbracket = \lambda x. |x| = 50$

Summarizing Landman's (2004) ontology of plurality, the property *fifty* holds of all plural individuals that have exactly fifty atomic members. Thus, *fifty lions* characterizes the set of lion pluralities with fifty atomic (lion) members. As desired, it is immediately false that *fifty lions* includes a plural individual containing exactly forty lions or exactly sixty lions. To arrive at the property denotation of numerals, Landman (2004) considers numerals to denote primitive numbers in the base case. That is, the numeral *fifty* denotes the number 50. Here, I do not directly follow Landman's (2004) exact path between type-*n* and type- $\langle e, t \rangle$  numerals, but the same general outcome holds: numerals behave like properties in constructions like (1) while they behave like numbers in constructions like (3). Here, it is not the case that *two* denotes a property of the baskets or of the mangos: (3) could be true if there were zero physical baskets involved in the purchase, and it is true if the speaker bought twenty (not two) mangos in total. Rather than denoting a property, *two* is a number-denoting argument to the unit *basketfuls*.

- (3) I bought two basketfuls (of mangos).

In the decades since its proposal, this dual view of numerals has taken hold in contemporary semantics, even among scholars with diverging viewpoints on the ontology of degrees (Champollion 2017, Scontras 2017, i.a.). In a given language, numerals need to behave as primitive number-denoting individuals in some structural configurations while they behave as properties of individuals in other configurations. This literature on numeral semantics, however, has largely taken the alternation between types *n* and  $\langle e, t \rangle$  to be a morphologically covert one. In English, for example, a numeral will be pre-nominal regardless of whether it is type-*n* or type- $\langle e, t \rangle$ , and its morphological form is similarly unaffected: the number *two* and the property *two* are both pre-nominal. Moreover, as summarized by Wagiel and Caha (2021) and Bylinina and Nouwen (2020), numeral theories have adopted various compositional pathways between types. As just two representatives from many in numeral semantics, Tatsumi (2025) takes numerals to have a basic type-*n* denotation, from which a type- $\langle e, t \rangle$  denotation is derived; meanwhile, Rothstein (2017) takes the reverse pathway.

Here, I broaden the study of numeral semantics to consider the case of Javanese, in which the overt morphosyntax of morphologically simple numerals flags whether the numeral is fitting a type-*n* or type- $\langle e, t \rangle$  slot in the composition. This paper's descriptive facts build on observations that date back in published work at least to Horne 1961 and Soemarmo 1979. Ishizuka 2008 contains important syntactic precursors to the present analysis, and the present paper represents

a significant expansion of [author’s own citation redacted for double-blind review]. I first refine these descriptions on simple numerals to build up complex numerals and establish the main claim: Javanese numerals cannot unilaterally have basic type- $n$  denotations. I do not refute claims that numerals can have a basic type- $n$  denotation; rather, Javanese presents a serious challenge to a theory of numerals that requires them all to have a basic type- $n$  denotation.

In this paper, I do not argue for a particular basic type of Javanese simple numerals; the challenges for a unilateral basic type- $n$  analysis of numerals stand independently.

## 1.2. Overview

In §2, I lay the empirical basis for the paper: for morphologically simple numerals, their overt position directly corresponds to their compositional role. I formalize the composition of DPs containing post-nominal and pre-unit simple numerals, and I identify a tension in complex numerals: they have the transparent morphosyntactic buildup of measure phrases like those meaning ‘three kilograms,’ but they have the semantic workload of simple numerals in being potential arguments of unit nouns. I argue in §3 that complex numerals are constituents in themselves. I then illustrate the main proposal for Javanese numeral composition, arguing that complex numerals cannot unilaterally have a basic type- $n$  denotation. I spend §4 rounding out the structure and composition of Javanese and the nominal phrases that host them, formally accounting for the linker suffix and offering two analyses with their implementations. I also consider homophonous linker morphology on numerals elsewhere in Austronesian languages. I conclude in §5. First, I provide background on Pemalang Javanese, the non-standard variety of Javanese represented in this paper.

## 1.3. Language background

The data in this paper come from the *kelurahan* ‘village’ called Mulyoharjo within the city of Pemalang, in the Pemalang *kabupaten* ‘regency,’ within the Indonesian province of Central Java. Javanese is characterized by immense dialectal variation; see [Conners 2008](#) for an overview of some of this variation and a grammar of one such dialect. To control for dialectal variation, all data reported here come from speakers who live within Mulyoharjo, which is approximately five square kilometers in area. Cognizant that there is variation within the city limits of Pemalang, I use the term ‘Pemalang Javanese’ in reference to Mulyoharjo Javanese in particular. There are approximately 1.5 million people who live in the Pemalang regency ([Badan Pusat Statistik 2025](#)), and Javanese is a common language in daily activities.

Pemalang Javanese is known for its marked departure from Standard Javanese, which is associated with the regions around Yogyakarta and Surakarta. To my knowledge, the distinctions are primarily phonological, phonetic, and lexical. For instance, word-final low vowels in Standard Javanese are pronounced as mid vowels in Pemalang Javanese; see [Isaura 2011](#) for a phonological characterization of Pemalang Javanese. The first-person singular pronoun *enyong* is a hallmark lexical item in Pemalang and other speech communities at the western edge of Central Java. As with Javanese in general, Pemalang Javanese includes distinct speech levels, the modulation of which is subject to complex sociolinguistic factors. All data here come from the *Ngoko* speech level. In broad strokes, *Ngoko* speech is typical of everyday speech between friends.

Unless otherwise cited, the data in this paper come from elicitations I conducted in-person and remotely with eleven native speakers of Pemalang Javanese between 2022 and 2025. Indonesian was used as the contact language. Data collection involved a series of judgement tasks, trans-

## 2. Setting a foundation: simple numerals and building MPs

In §2.1 I summarize the empirical distribution of nominal modifiers before contrasting pre-unit and post-nominal numerals: post-nominal numerals are fluidly ordered with respect to adjectives, all of which are post-nominal; pre-unit numerals are immediately before a noun that is construed as a measurement unit. I establish a compositional account of post-nominal numerals and pre-unit numerals in §2.2. In §2.3 I sketch the emergent puzzle of complex numerals. In that sketch, I show that the compositional account thus far under-generates: it wrongly predicts that complex numerals cannot themselves be arguments of unit nouns like *kilo* ‘kilogram.’

In this overview of Javanese DP structure, I annotate the expository examples with general part-of-speech tags: N(oun), Adj(ective), Adv(erb), Dem(onstrative), Num(eral), Q(uantifier).<sup>2</sup>

(4) Sing metu kuwe **kewan gedhe nemen**  
REL AV.emerge DEM animal large very  
'What emerges is a very large animal.'<sup>4</sup>

- Numerals also appear in the post-nominal field, as in (6). In (7), the post-nominal numeral *telu* appears after adjectives *lanang* ‘male’ and *cilik~cilik* ‘small.’

<sup>2</sup>For simplicity and space, I do not include examples with the syncretic possessive/definite marker *-e* on nominals also modified by numerals.

<sup>3</sup>I follow Leipzig glossing rules with the following additions: (N)AV for (non-)actor voice, LK for linker, NFML for the non-formal (*Ngoko*) speech register of Javanese, and REL for relativizer.

<sup>4</sup>The demonstrative in (4) is functioning in a copular clause, not as a direct modifier of the following nominal *kewan gedhe nemen*.

- (6) N Num  
 Ana **bocah telu** maran-i  
 exist child three AV.course-APPL  
 ‘Three children approach.’
- (7) N Adj Adj Num  
 Ana **bocah lanang cilik~cilik telu**  
 exist child male small~RED three  
 ‘There are three small boys.’

Moreover, post-nominal numerals are not strictly ordered with respect to other modifiers. Whereas the above examples show numerals as following the noun and adjectives, in (8) the numeral *loro* ‘two’ appears between the noun and the adjective *gedhe~gedhe* ‘big.’

- (8) N Num Adj  
 Sst ana **ula loro gedhe~gedhe** ka  
 shush exist snake two large~RED KA  
 ‘Shush, there are two big snakes.’ (PEM-130402)<sup>5</sup>

Numerals aside for a moment, quantifiers are the only modifiers that can linearly precede a Javanese DP’s head noun. For instance, *saben* ‘every’ in (9) and *kabeh* ‘all’ in (10) precede their respective lexical nouns being quantified over:

- (9) Q N  
 Grombyang-e ta’-pangan **saben dino**  
 k.o.stew-DEF 1SG-NAV.eat every day  
 ‘I eat the coconut-and-beef stew every day.’
- (10) Q N Dem  
 Enyong pesen **kabeh pangan-an kiye** kanggo enyong dhewek  
 1SG.NFML order all eat-NMLZ DEM for 1SG.NFML self  
 ‘I ordered all of this food for myself.’

The general order of DP-internal constituents is summarized in (11): in the post-nominal field, adjectives and numerals are not strictly ordered relative to each other. Foreshadowing the analysis of post-nominal numerals, post-nominal numerals can “mingle” (a term from [Landman 2004](#)) with other adjectives in Javanese, as they do within marked contexts in English as in (1).

- (11) A schematic of Javanese DP-internal ordering, not yet considering pre-unit numerals:  
 Q N { Adj, Num } Dem

Pre-unit numerals, unlike post-nominal numerals, appear with a suffix *-(a)ng* termed the linker by [Vander Klok and Matthewson \(2015\)](#) and others.<sup>6</sup> Pre-unit numerals obligatorily precede a unit; in the case of (12), this noun is *kilo* ‘kilogram.’

<sup>5</sup>Data with the “PEM”-prefixed citation come from [Connors and Sugianto 2011](#), a collection of transcribed narratives gathered in Pemalang.

<sup>6</sup>The realization of the numeral linker suffix on simple numerals is phonologically conditioned: *-ang* when the numeral is consonant-final (e.g., *pat-ang* ‘four’) and *-ng* when the numeral is vowel-final (e.g., *telu-ng* ‘three’). While there is variation in whether pre-unit numerals appear with the linker suffix—cosmopolitan Pemalang Javanese is less likely to pronounce the linker suffix than rural varieties—the starkest pattern is the unequivocal absence of the linker suffix on post-nominal numerals. The presence or absence of the linker on pre-unit numerals is not attributable to any difference in meaning. See §4.1 for further analysis of the linker.

- (12) Ana telu(-ng) kilo pelem  
 exist three-LK kilogram mango  
 ‘There are three kilograms of mangos.’
- (13) Ana pelem telu(\*-ng)  
 exist mango three  
 ‘There are three mangos.’

While the string *telu-ng kilo* ‘three kilograms’ can appear wholesale on either side of *pelem* ‘mango,’ its internal structure is fixed: the numeral must always precede the unit. The presence or absence of the linker does not salvage the violating permutations.

- (14) Possible and impossible permutations of the nominal ‘three kilograms of mangos’
- a. telu(-ng) kilo pelem  
 three-LK kilogram mango  
 ‘three kilograms of mangos’
  - b. pelem telu(-ng) kilo
  - c. \*telu(-ng) pelem kilo
  - d. \*pelem kilo telu(-ng)
  - e. \*kilo telu(-ng) pelem
  - f. \*kilo pelem telu(-ng)

Later in this paper, I account for the flexibility of *telu-ng kilo* in the extended nominal projection. My purpose in mentioning the two possible permutations in (14) is to motivate that calling *telu-ng* a “pre-nominal numeral” would be a misnomer: (14b) shows *telu-ng* linearly following the noun *pelem* ‘mango.’ For this reason, I instead call *telu-ng* a “pre-unit numeral.”

Pre-unit numerals are unambiguously associated with measurement interpretations, with important evidence coming from nouns that can be interpreted as containers or measurement units, such as *kranjang* ‘basket.’ On the noun itself, Javanese does not overtly mark a distinction between container and measurement-unit meanings. Rather, the numeral’s position and accompanying linker morphology flag the difference: a pre-unit, linker-marked numeral immediately preceding *kranjang* ‘basket’ is strictly associated with a ‘basketful’ interpretation. The construction in (15) refers to a volume of mangos.

- (15) Ana telu(-ng) kranjang pelem  
 exist three-LK basket  
 ‘There are three basketfuls of mangos.’

The pre-unit, linker-clad numeral *telu-ng* ‘three’ does not denote the number of baskets. Such a construction is felicitous in a context in which there are zero baskets:

- (16) Anyong weruh kire~kire telu-ng kranjang pelem nang ngarape anyong  
 1SG.NFML see guess~RED three-LK basket mango LOC in.front.of 1SG.NFML  
 ‘I see about three basketfuls of mangos in front of me.’

While pre-unit numerals give rise to measurement interpretations of the DP hosting them, post-nominal numerals can only have property denotations. The noun *kranjang* ‘basket’ is again

illustrative, this time in its container use. The post-nominal numeral construction is readily acceptable in the complete absence of mangos, as in (17). That is, the post-nominal numeral can have a property denotation, counting the baskets. Moreover, (18) indicates that the post-nominal numeral cannot be an argument to a measurement unit like ‘basketful’; the verb *ng-empet* ‘harvest’ selects direct objects that can be harvested, including fruits but excluding baskets. The post-nominal numeral can count containers but it cannot be used in an expression of measurement.

- (17) Enyong duwe **kranjang pelem loro** ning eseh kosong  
 1SG.NFML have basket mango two but still empty  
 ‘I have two mango baskets, but they are still empty.’

- (18) #Enyong wis ng-empet **kranjang pelem loro**  
 1SG.NFML already AV-harvest basket mango two  
 Lit.: ? ‘I have already harvested two mango containers.’

In summary, simple numerals’ overt positions map to their interpretation: pre-unit numerals are interpreted as numbers, while post-nominal numerals are interpreted as properties.<sup>7</sup>

## 2.2. Accounting for simple numerals

The main theoretical contribution of this paper—that numerals cannot unilaterally take type-*n* denotations as their basic denotations—is independent of the assumed basic type of simple numerals: *n* or  $\langle e, t \rangle$ . In §4, I consider possible implementations of both starting-points for simple numerals. For the rest of this section, I abstract away which is basic and which is derived to show that the problem of a unilateral basic type-*n* approach remains.

For consistency in the following data and derivations, I will primarily use the Javanese simple numeral *telu* ‘three.’ I will use **telu<sub>n</sub>** to refer to the number-denoting numeral ‘three’ and **telu<sub>\langle e, t \rangle</sub>** to refer to its property-denoting counterpart. I use  $|\dots|$  to denote the cardinality operation that takes an individual and returns the number of atomic parts of that individual.

- (20)  $\llbracket \mathbf{telu}_n \rrbracket = 3$

- (21)  $\llbracket \mathbf{telu}_{\langle e, t \rangle} \rrbracket = \lambda x. |x| = 3$

<sup>7</sup>For a restricted set of nouns, such as *basa* ‘language,’ counting requires that an immediately preceding linker-marked numeral be used rather than a post-nominal numeral. There are also some cases of pre-unit numerals that immediately precede a noun that is not readily interpretable as a measurement unit, such as *klambi* ‘shirt’ in the speaker-volunteered sentence in (19). Unlike *basa* ‘language’ and other nouns that cannot be counted using post-nominal numerals, *klambi* ‘shirt’ can also be modified by post-nominal numerals. While these cases of non-canonical pre-unit numerals are identified for upcoming work, preliminary observations and native speakers’ metalinguistic judgements suggest that these constructions are not interchangeable with their post-nominal numeral counterparts and instead are akin to group nouns like ‘a pair/trio of shirts.’

- (19) Enyong toko ro-ng klambi sisan  
 1SG.NFML buy two-LK shirt at.a.time  
 ‘I bought two shirts at a time.’



### 2.2.1. Post-nominal numerals denote properties

Recall that simple-numeral modifiers are obligatorily post-nominal. All adjectives in Javanese are beholden to this same post-nominal restriction (Vander Klok 2009). I consider **telu**<sub>(e,t)</sub>, like other nominal modifiers, to be obligatorily post-nominal.

In Javanese DPs at large, post-nominal numeral modifiers impose an interpretive restriction on the DP: the DP is an individual with as many atomic members as specified by the numeral. In (7), repeated in (22), the denotation of the existential pivot is the set of entities that are in the intersection of four sets: the set of children pluralities, the set of male entities, the set of contextually small entities (i.e., entities that are small for a child), and the set of plural entities with three atomic members. This target denotation of the DP, prior to existential closure, is summarized in (23).<sup>8</sup>

- (22)        N        Adj    Adj        Num  
 Ana **bocah lanang cilik~cilik telu**  
 exist child    male    small~RED three  
 ‘There are three small boys.’

- (23)  $\llbracket \text{bocah lanang cilik~cilik telu} \rrbracket = \lambda x. \text{CHILD}(x) \wedge \text{MALE}(x) \wedge \text{SMALL}(x) \wedge |x| = 3$

The goal denotation in (23) is readily available by a standard view of adjectival modification as set-intersection, as in Heim and Kratzer’s (1998) Predicate Modification. Whenever (22) is true, then it must be the case that there is a plural individual that is a member of the four sets mentioned above: the set of children, the set of male entities, the set of small entities (or entities that are small for a child), and the set of entities that each have three atomic parts.

Characterizing the set of entities that each have three atomic parts, the property-denoting **telu**<sub>(e,t)</sub> composes with the other three properties by Predicate Modification; it also yields the desired syntactic category for the attested word-order patterns. Post-nominal numerals are not strictly ordered with respect to other adjectives, but they cannot precede the noun itself.

- (24)    a.  $\llbracket \text{bocah} \rrbracket = \lambda x. \text{CHILD}(x)$   
           b.  $\llbracket \text{lanang} \rrbracket = \lambda x. \text{MALE}(x)$   
           c.  $\llbracket \text{cilik~cilik} \rrbracket = \lambda x. \text{SMALL}(x)$

Post-nominal numerals, like other modifiers in Javanese, are adjoined to the NP. For contrast with the upcoming treatment of numerals in measurement constructions, I show the NP structure for *pelem telu* ‘three mangos.’ I take simple numerals to be phrasal; see §4.1 for evidence from *wh*-replacement.

- (25)
- |                                                                      |                                                                   |
|----------------------------------------------------------------------|-------------------------------------------------------------------|
| NP                                                                   |                                                                   |
| $\lambda x. \text{MANGO}(x) \wedge  x =3$                            |                                                                   |
| NP                                                                   | NumP                                                              |
| $\triangle$<br><i>pelem</i><br>mango<br>$\lambda x. \text{MANGO}(x)$ | $\triangle$<br><i>telu</i> <sub>(e,t)</sub><br>$\lambda x.  x =3$ |

<sup>8</sup>Javanese nouns are number-neutral, and I make the non-crucial assumption in this paper that morphologically bare nouns in Javanese range over atomic and plural individuals. The reduplication of *cilik* ‘small’ in (22) may be a reflex of plurality or pluractionality, similar to reduplication patterns in Balinese (Arka and Dalrymple 2017).



### 2.2.2. Pre-unit numerals denote numbers

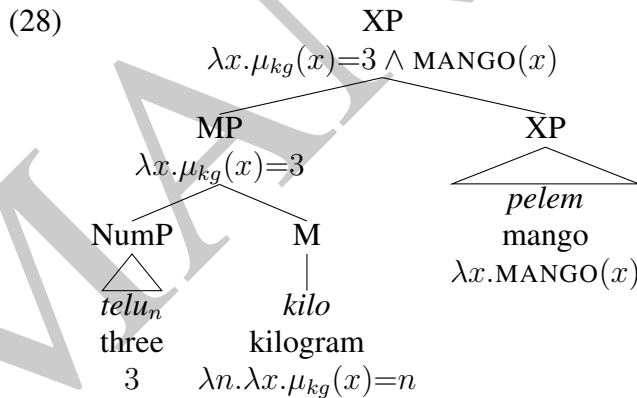
The pre-unit, linker-clad numeral *telu-ng* ‘three’ in (12), repeated in (26), does not denote the number of mangos nor does it denote the number of physical kilograms—the latter case is uninterpretable in any context since literal kilograms (i.e., kilograms outside of a measurement-unit construal) cannot be counted. While post-nominal numerals are intersective modifiers, pre-unit numerals are not. The numeral *telu-ng* in (26) is not specifying the cardinality of the mangos but rather the number of kilograms that the mangos measure.

- (26) Ana *telu(-ng) kilo pelem*  
 exist three-LK kilogram mango  
 ‘There are three kilograms of mangos.’

To compose the nominal *telu-ng kilo pelem* ‘three kilograms of mangos,’ I assume that measurement units like *kilo* ‘kilogram’ are functions from numbers to properties of entities, following Rothstein 2017. In this section, I do not represent the linker in the denotations; I revisit the role of the linker in §4.1. Taking the numeral *telu* ‘three’ as  $\mathbf{telu}_n$  in its type- $n$  denotation, Functional Application produces the intermediate result in (27c) of an  $\langle e, t \rangle$  property: the property of measuring three kilograms. Predicate Modification ensues, such that the resulting phrase characterizes the set of mango individuals that measure three kilograms.

- (27) a.  $\llbracket \mathbf{kilo} \rrbracket = \lambda n. \lambda x. \mu_{kg}(x) = n$   
 b.  $\llbracket \mathbf{telu}_n \rrbracket = 3$   
 c.  $\llbracket \mathbf{kilo} \rrbracket(\llbracket \mathbf{telu}_n \rrbracket) = \lambda x. \mu_{kg}(x) = 3$   
 d.  $\llbracket \mathbf{pelem} \rrbracket = \text{MANGO}(x)$   
 e.  $\llbracket \mathbf{telu kilo pelem} \rrbracket = \lambda x. \mu_{kg}(x) = 3 \wedge \text{MANGO}(x)$

Syntactically, I consider phrases like *telu-ng kilo* ‘three kilograms’ to be measure phrases (MPs) that adjoin to the nominal projection. For now, I label the MP’s adjunction site as XP to remain agnostic to the particular level of structure projected. The level of structure must be high enough to accommodate demonstrative- or determiner-marked phrases that MPs then adjoin to, as I show in §4.3. For that reason, I assume XP to be higher than NP.



### 2.3. The emergent puzzle: complex numerals

By the account developed so far, there are two compositional slots for morphologically simple numerals: a slot immediately preceding a unit noun, for type- $n$  numbers; and a slot in the post-

nominal field, for type- $\langle e, t \rangle$  properties. Complex numerals show that the machinery developed until now is insufficient.

Complex numerals contain readily identifiable subparts. Minimally, a complex numeral will include a simple numeral (a numeral ‘one’ through ‘nine’) as the linker-marked multiplier, followed by a counting base. For instance, (29) showcases complex numerals that all use the simple numeral *telu-ng* ‘three’ as the multiplier and various counting bases: *puluh* ‘unit of ten,’ *lawe* ‘unit of twenty-five,’ and *atus* ‘unit of hundred.’

- (29) a. *telu-ng puluh*  
three-LK unit.of.ten  
‘thirty’  
b. *telu-ng lawe*  
telu-LK unit.of.25.NFML  
‘seventy five’  
c. *telu-ng atus*  
three-LK unit.of.hundred  
‘three hundred’

Counting bases cannot stand alone to denote the cardinality of a set. In this respect, a counting base like *puluh* ‘unit of ten’ is more parallel to the distribution of English *dozen* than that of English *ten*. In general, counting bases like *puluh* never appear without a preceding simple numeral functioning as the multiplier.

- (30) Ana bocah {*se-puluh/* \* $\emptyset$ -*puluh*}  
exist child one.LK-unit.of.ten  
‘There are ten children.’

Within a complex numeral, the morphosyntax is immediately reminiscent of the MPs discussed starting in §2.1: a pre-unit numeral linearly precedes a measurement unit, and the pre-unit numeral can take the linker suffix. Examining the phrases in isolation, the only distinction between *telu-ng kilo* ‘three kilograms’ and *telu-ng puluh* ‘thirty’ (lit. ‘three tens’) is that the former uses a measurement unit of kilograms while the latter uses a measurement unit of tens. Situating these phrases in full clauses will problematize this parallel between MPs like *telu-ng kilo* and complex numerals like *telu-ng puluh*. In brief, complex numerals have the morphosyntactic buildup and syntactic distribution of MPs, but they can carry out the semantic work of simple numerals in being eligible arguments to unit nouns.

In (27), I treated an MP like *telu-ng kilo* ‘three kilograms’ as an  $\langle e, t \rangle$  function. It is empirically motivated to treat complex numerals like *se-puluh* ‘ten’ or *telu-ng puluh* ‘thirty’ as  $\langle e, t \rangle$  functions, too. Besides the morphological similarity to MPs of physical measurement, complex numerals can readily precede the nouns that they count over—in marked contrast with obligatorily post-nominal simple numerals like *loro* ‘two’ as shown in (31a).

- (31) a. Tono toko {*se-puluh/* \**loro*} kursi  
Tono buy one.LK-ten two chair  
OK: ‘Tono bought ten chairs.’  
Unavailable here: ‘Tono bought two chairs.’  
b. Tono toko kursi {*se-puluh/* *loro*}  
Tono buy chair one.LK-unit.of.ten two  
‘Tono bought ten/two chairs.’

In §4.3, I return to account for the post-nominal position available to complex numerals as in (31b). For now, I note that the apparent flexibility of complex numerals is further evidence to represent them on par with MPs like *telu-ng kilo* ‘three kilograms’: such MPs can likewise appear wholesale on either side of the noun being measured, as in (14). For now, I take complex numerals to be  $\langle e, t \rangle$  functions, just as MPs in general are.

Taken together, the two types of simple complex numerals and the basic- $\langle e, t \rangle$  denotation of complex numerals neatly accounts for the data thus far: of simple numerals as modifiers, of simple numerals as arguments in MPs, and of complex numerals as modifiers. The proposal then predicts a type clash when a type- $\langle e, t \rangle$  complex numeral like *telu-ng puluh* ‘thirty’ is supplied as an argument to a  $\langle n, \langle e, t \rangle \rangle$  function like *kilo* ‘kilogram.’ This prediction is directly falsified: *telu-ng puluh* can be such an argument for *kilo*.

- (32) Ana telu-ng puluh kilo pelem  
 exist three-LK unit.of.ten kilogram mango  
 ‘There are thirty kilograms of mangos.’

The remainder of the paper is concerned with analyzing complex numerals such as *telu-ng puluh* ‘thirty’: they are transparently built like type- $\langle e, t \rangle$  MPs such as *telu-ng kilo* ‘three kilograms’ but they fit the type- $n$  slot for number-denoting arguments of unit nouns. Under the assumption that complex numerals are built to be  $\langle e, t \rangle$  properties, just as MPs are, then complex numerals, then, require a second life of their own. That is, complex numerals shuttle from a basic type- $\langle e, t \rangle$  denotation to a derived type- $n$  denotation. To perform this shuttling, I employ an operator called CARD following [Bylinina and Nouwen 2020](#), for complex numerals. Cross-linguistically, the Javanese findings are informative in what numeral semantics *cannot* do: universally endow numerals with basic type  $n$ .

### 3. Javanese complex numerals

In this section, I defend that (32) indeed has *telu-ng puluh* as a constituent. This defence is necessary, since any analysis of complex numerals’ type requires them to be constituents of their own in the crucial data like (32). With these constituency facts in hand, I argue that unilaterally giving numerals a basic number denotation runs into empirical challenges.

#### 3.1. Complex numerals are constituents

##### 3.1.1. Two competing analyses of complex numerals

In Javanese, complex numerals like *telu-ng puluh* ‘thirty’ are constituents. The constituency of complex numerals, not taken for granted cross-linguistically ([Ionin and Matushansky 2018](#), [Tatsumi 2025](#)), legitimizes the puzzle of how complex numerals can fit into type- $n$  slots and type- $\langle e, t \rangle$  slots. I will show that the constituency in (33) is the correct constituency analysis. I note here that the linker suffix cannot attach to the right edge of complex-numeral argument of a unit noun.

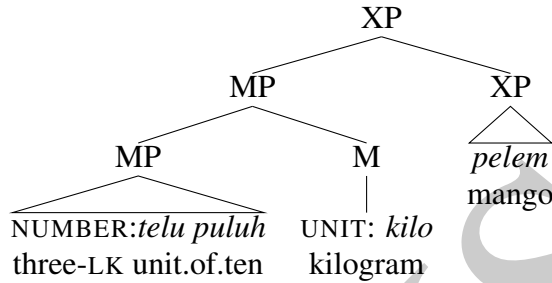
- (33) Ana [MP-1[MP-2 telu-ng puluh](\*-ng) kilo] pelem  
 exist three-LK unit.of.ten kilogram mango  
 ‘There are thirty kilograms of mangos.’

Consider an alternative constituency analysis of (33), in which *telu-ng* ‘three’ and *puluh* ‘unit of ten’ do not form a constituent. Assuming a binary-branching structure and that the entire phrase *telu-ng puluh kilo* ‘thirty kilograms’ is a constituent, then the unique alternative analysis is the one in which *puluh kilo* is a constituent instead as in (34):

- (34) Ana [<sub>MP</sub> *telu-ng* [<sub>puluh(\*-ng)</sub> *kilo*]] *pelem*  
 exist three-LK unit.of.ten kilogram mango  
 ‘There are thirty kilograms of mangos.’

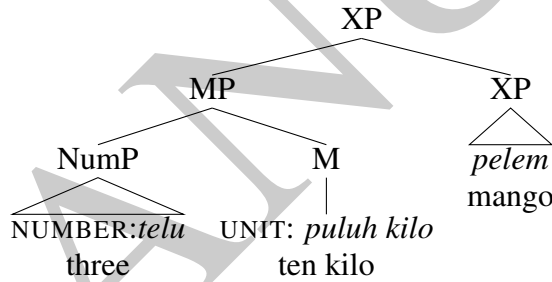
In comparing these two constituency analyses, I will use the terms SIMPLE-UNIT for the analysis in (33) and SIMPLE-NUMBER for the analysis in (34). In the SIMPLE-UNIT hypothesis, the measurement unit is just *kilo* ‘kilogram’; the complexity is instead loaded into the NUMBER slot, which holds not just a simple numeral *telu-ng* but also the counting base *puluh*. The tree in (35) summarizes the SIMPLE-UNIT view.

- (35) SIMPLE-UNIT hypothesis: complex NUMBER, simple UNIT



The SIMPLE-NUMBER hypothesis, in contrast, reserves the NUMBER slot only for the simple numeral *telu-ng*. Complexity is loaded into the unit slot, which hosts the counting base *puluh* and the physical measurement unit *kilo*. The SIMPLE-NUMBER view is illustrated in (36).

- (36) SIMPLE-NUMBER hypothesis: simple NUMBER, complex UNIT



An immediate appeal of the SIMPLE-NUMBER hypothesis is that the distribution of the linker suffix becomes less mysterious: The linker only attaches to number-denoting arguments of unit nouns: morphologically simple numerals like *telu* ‘three’ are thus eligible, while complex numerals like *telu-ng puluh* ‘thirty’ are not. Furthermore, the SIMPLE-NUMBER hypothesis illustrated in (36) would eliminate the puzzle of complex numerals’ dual semantic lives. By this approach, complex numerals do not have any semantic type to begin with because they are not constituents. Under this approach, the NUMBER slot only takes type-*n* simple numerals.

Despite its initial draw, the SIMPLE-NUMBER hypothesis should ultimately be set aside in favour of the SIMPLE-UNIT hypothesis. Evidence from approximatives and disjunction support the conclusion that complex numerals like *telu-ng puluh* ‘thirty’ are constituents in Javanese. If

that conclusion must be true, then the SIMPLE-NUMBER hypothesis cannot be correct. Next, I summarize the results and their implications for each hypothesis. While developed independently of [Tatsumi 2025](#), the diagnostic tests are similar and yield similar results about complex numerals in typologically distinct languages: complex numerals can be constituents.

### 3.1.2. Coordination

The principal difference between the SIMPLE-NUMBER and SIMPLE-UNIT hypotheses is whether a complex numeral is a constituent: the SIMPLE-NUMBER hypothesis predicts no; the SIMPLE-UNIT hypothesis predicts yes. Here, I show that a complex-numeral constituents like *telu-ng puluh* ‘thirty’ can be coordinated, as predicted by the SIMPLE-UNIT hypothesis. Meanwhile strings analyzed as complex-unit constituents under the SIMPLE-NUMBER cannot be coordinated.

First, I show that complex numerals can be coordinated.<sup>9</sup> The forms *telu-ng puluh* ‘thirty’ and *pat-ang puluh* ‘forty’ can be coordinated in a disjunctive question.

- (38) % Context: Your friend told you that she bought mangos. You’re certain that she told you one of two quantities of mangos purchased: either thirty kilograms or forty kilograms.

Apa ana telu-ng puluh utawa pat-ang puluh kilo pelem?

Q exist three-LK ten or four-LK ten kilogram mango

‘Were there thirty or forty kilograms of mangos?’

While the physical measurement unit *kilo* can be shared in the above disjunction, the counting base *puluh* cannot be shared by both simple-numeral multipliers. The only available reading is an interpretation of *puluh* strictly in the second disjunct. The set of alternatives in (39) exhaustively includes ‘three kilograms of mangos’ and ‘forty kilograms of mangos.’

- (39) Apa ana telu-ng utawa pat-ang puluh kilo pelem?

Q exist three-LK or four-LK ten kilogram mango

Unavailable: ‘Were there thirty or forty kilograms of mangos?’

OK: ‘Were there three or forty kilograms of mangos?’

In (39), the simple numeral *telu-ng* ‘three’ and complex numeral *pat-ang puluh* ‘forty’ are doing the same semantic work—specifying a number of kilogram units—and the coordinated syntax affirms that these elements are filling the same syntactic position.

### 3.1.3. Approximative constructions

The next Javanese diagnostic follows an argument summarized in [Jiang et al. 2022](#) for numeral-containing phrases in Mandarin: approximative marking flags the constituency of a complex numeral. As in Mandarin, in Javanese the attachment site of approximative morphemes and their

<sup>9</sup>I focus this portion of coordination data on disjunction because conjunction introduces a confounding factor: conjunction is the means of making complex numerals that are not exact multiples of the counting bases. For instance, (37) uses conjunction of the numerals *telu-ng puluh* ‘thirty’ and *lima* ‘five’ to form the numeral ‘thirty-five.’

- (37) telu-ng puluh lima pelem  
three-LK unit.of.ten five mango  
‘thirty-five mangos’

subsequent interpretation—namely, the locus of approximation—is readily compatible with the SIMPLE-UNIT hypothesis: complex numerals are constituents. This result adds to the mounting problems for the SIMPLE-NUMBER hypothesis.

The multifarious suffix *-an* can attach to complex numerals to convey imprecision or uncertainty (Robson 2014),<sup>10</sup> often but not necessarily in concert with other markers like the adverbial *kire~kire* ‘to my best guess.’ Complex numerals inside of MPs can bear this suffix, as *telu-ng puluh* ‘thirty’ does in (42). Complex numerals that directly count the lexical noun can also be modified by *-an*, as shown in (43).

- (42) Ana kire~kire telu-ng puluh-an kilo pelem  
 exist think~RED three-LK ten-APPROX kilogram mango  
 ‘There are about thirty-ish kilograms of mangos. (Maybe twenty-eight, for instance.)’
- (43) Ana kire~kire se-puluh-an pelem  
 exist think~RED one.LK-ten-APPROX mango  
 ‘There are about ten-ish mangos. (Maybe nine, maybe eleven, ...)’

As the parenthetical phrases suggest, the locus of approximation is provided by the complex numeral itself: in the case of (42), the estimated volume of mangos is thirty. Judging by linear order alone, a tempting alternative is that *-an* directly composes with the counting base—that is, *puluh* ‘unit of ten’ in both examples above. Under that analysis, the uncertainty is about the counting base. Applying this analysis to (42), the sentence would be expected to convey uncertainty whether the volume of mangos is thirty kilograms (three tens), three kilograms (three ones), or 75 kilograms (three twenty-fives); the closest available counting bases to *puluh* ‘unit of ten’ are ones and *lawe* ‘unit of twenty five,’ the latter exemplified in (29). More dramatically, this “low-attachment” analysis of *-an* would incorrectly predict that (43) can be paraphrased with the following: “There is a number of mangos. Maybe that number is ten, maybe one. I’m not sure of the counting base, but whatever the counting base is, its multiple is one.” The meaning is unattested. Rather than the immediately adjacent morpheme that serves as the counting bases (e.g., *puluh* ‘unit of ten’), the reported locus of approximation is the complex numeral as a whole (*se-puluh* ‘ten’).

Corroborating the coordination data, the morphosyntax of approximation indicates that complex numerals are syntactic and semantic constituents in Javanese. The final piece of evidence in favour of the SIMPLE-UNIT hypothesis is indirect: the SIMPLE-NUMBER hypothesis poses a challenge for composition.

### 3.1.4. Composition inside a complex unit

Suppose that the SIMPLE-NUMBER hypothesis were true. Then a complex numeral would not be a constituent inside of a larger MP. Made concrete, the constituency in (44) would hold: *telu-ng* ‘three’ does not form a constituent with *puluh* ‘unit of ten’ to the exclusion of *kilo* ‘kilogram.’

<sup>10</sup>In attaching to numerals, the suffix *-an* can express approximation as well as distributivity: contrast (40) with (41). In the former, the number of breads is equal to the number of children; in the latter, there is one bread shared among the children.

- (40) Bocah~bocah oleh roti siji-nan  
 child~RED get bread one-DIST  
 ‘The children each got a bread.’

- (41) Bocah~bocah oleh roti siji  
 child~RED get bread one  
 ‘The children got a bread.’

- (44) Ana [<sub>MP</sub> *telu-ng* [*puluh kilo*]] *pelem*  
 exist three-LK unit.of.ten kilogram mango  
 ‘There are thirty kilograms of mangos.’

There are two functions that take number-denoting arguments that must compose: *puluh* ‘unit of ten’ and *kilo* ‘kilogram.’ To be as formally charitable as possible for this opposing analysis, I will allow that Predicate Modification could be generalized to allow any two property-denoting expressions, not just  $\langle e, t \rangle$  functions. This generalization will not be enough to salvage this opposing analysis.

- (45) a.  $\llbracket \text{puluh} \rrbracket = \lambda n. \lambda x. \mu_{\text{ten}}(x) = n$   
 b.  $\llbracket \text{kilo} \rrbracket = \lambda n. \lambda x. \mu_{\text{kg}}(x) = n$   
 c. Assuming generalized Predicate Modification:  
 $\llbracket \text{puluh kilo} \rrbracket = \lambda n. \lambda x. \mu_{\text{ten}}(x) = n \wedge \mu_{\text{kg}}(x) = n$

Even if this generalization of Predicate Modification held, the derivation makes incorrect predictions: in that case *telu-ng* [*puluh kilo*] would be predicted to deliver the conjoined property of measuring three tens and of measuring three kilograms. Then (44) would be incorrectly predicted to be true when there exists a plurality of mangos that measure three kilograms and that have thirty atomic parts. (44) is true if there exists a plurality of mangos that cumulatively measure thirty kilograms, regardless of the number of atomic parts. The prediction in (45c) would incorrectly impose a cardinality-30 requirement and a 3-kg requirement on the set it characterizes.

Taken alongside the arguments earlier in this section, I conclude that complex numerals form constituents of their own. The following constituency holds, where *telu-ng puluh* is a constituent.

- (46) Ana [<sub>MP-1</sub> [<sub>MP-2</sub> *telu-ng puluh*](*\*-ng*) *kilo*] *pelem*  
 exist three-LK unit.of.ten kilogram mango  
 ‘There are thirty kilograms of mangos.’

### 3.2. Formalizing CARD

In this section, I have motivated that Javanese complex numerals are constituents, including when they are interpreted as arguments of a unit noun. The analysis must enable complex numerals to fit the type-*n* slot reserved for a unit noun’s argument. In this section, I augment the account in §2 by introducing the CARD operator to compose with type- $\langle e, t \rangle$  complex numerals to deliver a type-*n* argument for input to unit nouns like *kilo* ‘kilogram.’ I begin by showing the success of beginning with a basic property denotation of complex numerals. In §3.4 I rule out an alternative analysis.

Bylinina and Nouwen (2020) introduce the CARD operator, which I call CARD<sub>OLD</sub> in (47) to foreshadow that I slightly revise their denotation in this section.

- (47)  $\llbracket \text{CARD}_{\text{OLD}} \rrbracket = \lambda P_{\langle e, t \rangle}. \iota n. \forall x [P(x) \rightarrow |x| = n]$

A type- $\langle e, t \rangle$  complex numeral like *telu-ng puluh* ‘thirty’ can compose with CARD<sub>OLD</sub> to return the unique number *n* such that all entities in with the property of measuring three tens have cardinality *n*. In the case of *telu-ng puluh* ‘thirty,’ this *n* is 30.

- (48) a.  $\llbracket \text{telu puluh} \rrbracket = \lambda x. \mu_{\text{ten}}(x) = 3$



$$b. \llbracket \text{CARD}_{\text{OLD}} \rrbracket(\llbracket \text{telu-ng puluh} \rrbracket) = 30$$

At first blush,  $\text{CARD}_{\text{OLD}}$  is a success in accounting for the Javanese numeral data: it shuttles complex numerals from a basic type- $\langle e, t \rangle$  denotation to a type- $n$  denotation that is needed when complex numerals are themselves arguments of unit nouns like *kilo* ‘kilogram.’ However, the denotation of this operator must be revised.

In its formulation in (47),  $\text{CARD}_{\text{OLD}}$  takes any  $\langle e, t \rangle$  function as its input. Recall that *telu-ng kilo* ‘three kilograms’ is treated as an  $\langle e, t \rangle$  function, as is *pelem telu* ‘three mangos.’ Each of these inputs pose challenges for  $\text{CARD}_{\text{OLD}}$ . First, in the case of ‘three kilograms’: there is no unique number  $n$  such that all 3-kg entities have cardinality  $n$ . Consider, for instance, the cardinality of a plurality of paperclips measuring 3 kilograms versus the cardinality of a plurality of mangos measuring 3 kilograms. The denotation in  $\text{CARD}_{\text{OLD}}$  must be revised in its domain restriction, to ensure the existence and uniqueness of this number  $n$ .

Second, in the case of ‘three mangos’: any element of the set of three-mango pluralities will have cardinality 3. In this respect,  $\text{CARD}_{\text{OLD}}$  is incorrectly predicted to combine with *pelem telu* ‘three mangos’ to return the number 3, which could then be a number-denoting argument for a unit noun like *kilo* ‘kilogram.’ However *[pelem telu] kilo* is ill-formed and cannot mean ‘three kilograms’ as this naïve implementation of  $\text{CARD}_{\text{OLD}}$  would predict.

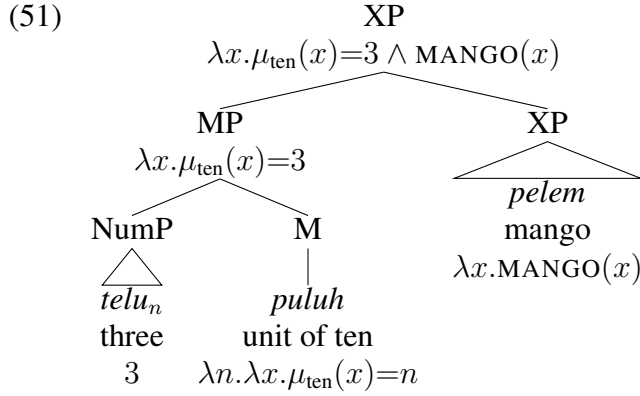
- (49) a.  $\llbracket \text{pelem telu} \rrbracket = \lambda x. \text{MANGO}(x)$   
 b.  $\llbracket \text{CARD}_{\text{OLD}} \rrbracket(\llbracket \text{pelem telu} \rrbracket) = 3$

Both of these challenges can be resolved by restricting the domain of the operator. Henceforth, I use  $\text{CARD}_{\text{NEW}}$  to refer to the updated denotation of the operator as in (50). In this updated denotation, I restrict the domain to  $\langle e, t \rangle$  functions  $P$  that have exactly one number  $n$  such that an entity has cardinality  $n$  if and only if that entity is in the set characterized by  $P$ . The existence and uniqueness requirements resolve the first challenge, posed by *telu-ng kilo* ‘three kilograms.’ The biconditional resolves the second challenge, posed by *pelem telu* ‘three mangos.’ As a result,  $\text{CARD}_{\text{NEW}}$  can only compose with  $\langle e, t \rangle$  functions that are numerals, as desired.

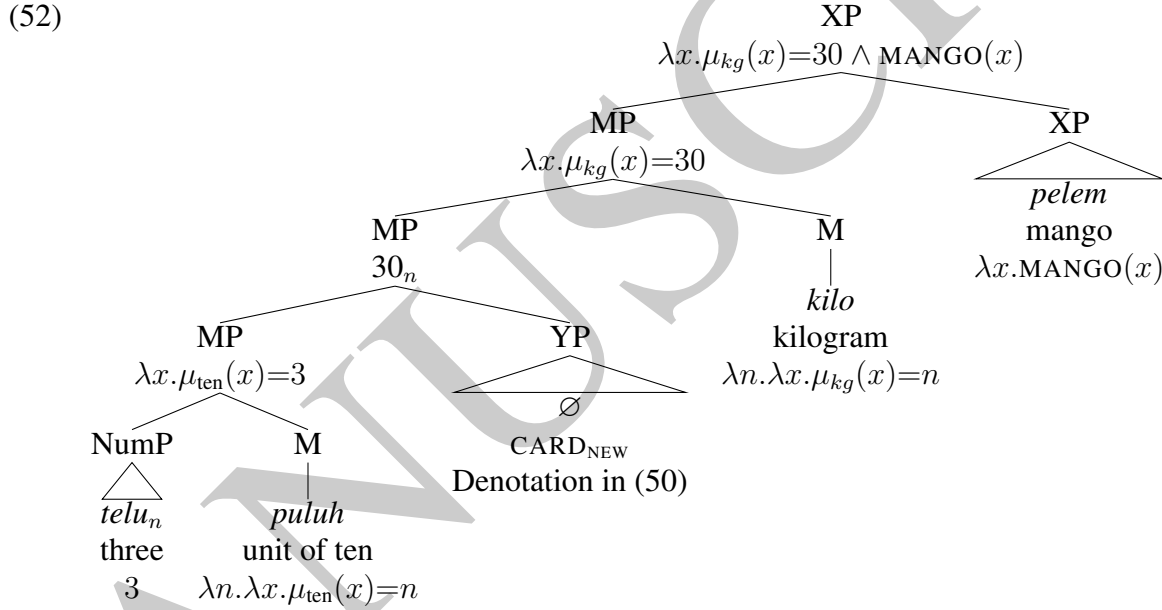
$$(50) \quad \llbracket \text{CARD}_{\text{NEW}} \rrbracket = \lambda P_{\langle e, t \rangle} : \exists! n \forall x [P(x) \leftrightarrow |x| = n]. \iota n. \forall x [P(x) \rightarrow |x| = n]$$

### 3.3. $\text{CARD}_{\text{NEW}}$ at work: composing complex numerals

Here, I show the composition of complex numerals as number-denoting arguments and as property-denoting modifiers. Beginning with complex numerals’ basic type- $\langle e, t \rangle$  denotation, I consider the complex numeral *telu-ng puluh* ‘thirty’ as a modifier. Inside the complex numeral, I continue with the assumption from §2 that the simple numeral *telu* ‘three’ can somehow obtain a type- $n$  denotation, either by its basic or derived type. After Functional Application composes the simple numeral and the unit noun, Predicate Modification between the MP and the substance *pelem* ‘mango’ provides the desired denotation of the entire nominal: the set of entities that measure three tens and that are mangos.



Extending this system to MPs that themselves contain complex numerals,  $\text{CARD}_{\text{NEW}}$  enters the composition. Supplied with the complex numeral *telu-ng puluh* ‘thirty,’  $\text{CARD}_{\text{NEW}}$  returns the number 30 for subsequent input to the unit noun *kilo* ‘kilogram.’ Again, Predicate Modification at the top node of (52) achieves the desired denotation: the set of entities that are mangos and that measure 30 kilograms.



With the introduction of  $\text{CARD}_{\text{NEW}}$ , the present account is able to cover the type- $n$  and type- $\langle e, t \rangle$  denotations of both simple and complex numerals. Up until this point, I have abstracted away the pathway taken by simple numerals: I have not committed to a basic type for numerals like *telu* ‘three.’ In the next section, I explore two analyses. Across both, the main theoretical contribution remains: unilaterally treating all Javanese numerals with basic type- $n$  denotations proves problematic, as I show now.

### 3.4. Ruling out a unilateral type- $n$ starting point

Here, I argue against an analysis that gives all Javanese numerals a basic type  $n$ . My argument is a proof by contradiction. Assume that all numerals have basic type  $n$ . Then there is a choice

point: there is a compositional rule of Degree Modification as in (53),<sup>11</sup> a syntactically restricted operator like MANY (54), or neither in the grammar. Pursuing any of these options will quickly lead to challenges.

- (53) **Degree Modification** If  $\alpha$  has exactly two daughters,  $\beta$  and  $\gamma$ , such that  $\llbracket \beta \rrbracket \in D_d$  and  $\llbracket \gamma \rrbracket \in D_{et}$ , then  
 $\llbracket \alpha \rrbracket = \lambda x_e. \#(x) = \llbracket \beta \rrbracket(x) \wedge \llbracket \gamma \rrbracket(x)$
- (54)  $\llbracket \text{MANY} \rrbracket = \lambda n. \lambda x. |x| = n$

**Case 1: Neither Degree Modification nor MANY are available.** Functional Application and Predicate Modification alone cannot compose type- $n$  simple numerals with  $\langle e, t \rangle$  predicates like *kursi* ‘chair.’ However, such modifications are readily available:

- (55) Tono toko kursi {se-puluh/            loro}  
 Tono buy chair one.LK-unit.of.ten two  
 ‘Tono bought ten/two chairs.’

As a result, this case under-generates.

**Case 2: Composition with Degree Modification.** With Degree Modification, then all numerals can compose directly with predicates like *kursi* ‘chair’ by a syntactically unrestricted rule of composition. However, while simple and complex numerals can follow the lexical noun, only complex numerals can occur before the noun:

- (56) Tono toko {se-puluh/ \*loro } kursi  
 Tono buy one.LK-ten two chair  
 OK: ‘Tono bought ten chairs.’  
 Unavailable here: ‘Tono bought two chairs.’

Without further stipulation, simple numerals would be incorrectly predicted to have the same word-order patterns as complex numerals do. This case over-generates.

**Case 3: Composition with MANY.** If we assume that MANY is a covert adjective, then like all adjectives in Javanese MANY is post-nominal. However, without some new (and also silent) MANY<sub>2</sub> that only takes complex numerals, this account would wrongly predict that complex numerals can only linearly follow lexical nouns being counted over. Shy of asserting this separate MANY<sub>2</sub> operator, which also must be covert but instead only combine with complex numerals, this case under-generates.

Javanese numerals pose challenges for accounts that would unilaterally endow simple and complex numerals alike with a basic type- $n$  denotation. The following section offers two solutions: (i) simple numerals having basic type- $n$  denotations and complex numerals having a basic type- $\langle e, t \rangle$  denotations, and (ii) both simple and complex numerals having basic  $\langle e, t \rangle$  denotations.

<sup>11</sup>In this usage of Degree Modification, I take numbers to be a subset of  $D_d$ ; see §4.4 for a discussion of the relationship between number and degree.

## 4. Implementation and discussion

### 4.1. The linker's sensitivity to hierarchical structure

Throughout this paper, I have refrained from giving a denotation to the numeral linker suffix, realized as *-ng* or *-ang* on simple pre-unit numerals. Here, I show that the linker must be sensitive to hierarchical syntactic structure.

Recall that the linker is not always present when it attaches to simple numerals. Its absence on nasal-final simple numerals like *enem* ‘six’ is readily accounted for under a basic phonological analysis. What is more troubling for this kind of contextual-allomorphy account is the categorical restriction against linkers on the right edge of complex-numeral arguments, as in (57). If the linker were only sensitive to its linear phonological environment, the final phonological segment of the complex numeral *telu-ng ewu* ‘three thousand’ is wrongly predicted to be a well-formed linker attachment site. Recall that the linker readily suffixes to the simple numeral *telu* ‘three.’

- (57) Ana [[ *telu-ng ewu*](*\*-ng*)      *kilo*]      *pelem*  
 exist three-LK unit.of.thousand kilogram mango  
 ‘There are three thousand kilograms of mangos.’

The linker is sensitive to hierarchical structure, not linear order. Numerals that are not purely multiplicative, like *telu-ng puluh lima* ‘thirty-five,’ are built by null coordination of a complex numeral and a simple numeral. When such a numeral is the argument of a unit noun like *kilo*, a right-edge linker is unavailable as in (58a), a restriction that is absent on a standalone simple-numeral argument as in (58b).

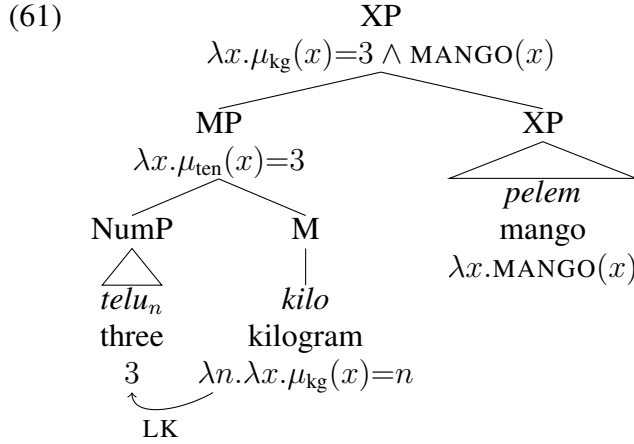
- (58) a. Ana *telu-ng puluh lima*(*\*-ng*) *kilo*      *pelem*  
 exist three-LK unit.of.ten five      kilogram mango  
 ‘There are thirty-five kilograms of mangos.’  
 b. Ana *lima-ng kilo*      *pelem*  
 exist five-LK kilogram mango  
 ‘There are five kilograms of mangos.’

In this section, I treat the linker as a byproduct of syntactic licensing. Namely, the linker is the result of licensing by the unit noun combining with a simple numeral. I take linker-marking to be the result of a Spec-Head Agree relation, with simple numerals being phrasal<sup>12</sup> and in the specifier position of the MP projected by the unit noun as in (61).

<sup>12</sup>Following cross-linguistic observations in [Tatsumi 2025](#), suggestive evidence for the phrasal status of simple numerals within MPs comes from *wh*-replacement. The *wh*-word *pira* ‘how many’ directly parallels simple numerals in its morphology and syntactic position: in a question of cardinality, *pira* is post-nominal and linker-less (59); in a question of measurement, *pira* is pre-unit and linker-clad (60).

- (59) Ana *pelem pira*(*\*-ng*)?  
 exist mango how.many  
 ‘How many mangos are there?’

- (60) Ana *pira-ng kilo*      *pelem*?  
 exist how.many-LK kilogram mango  
 ‘How many kg. of mangos are there?’



Foreshadowing the possible analyses of simple numerals in the next subsection, the lack of linkers on complex numerals could be attributable to one of two factors: (i) a purely syntactic-category account, by which complex numerals are not NumPs and thus invisible to the Agree relation associated with the linker; or (ii) an appeal to the  $\text{CARD}_{\text{NEW}}$  operator as denoted in (50) syntactically intervening between the complex numeral and the unit noun. Note that  $\text{CARD}_{\text{NEW}}$  enters the composition at the exact point where the linker is barred on complex numerals.

While the linker has been mentioned in descriptive and analytical work on Javanese, its semantic import has not been specified, vacuous or otherwise. Phonetically similar linker suffixes are prevalent across Austronesian languages (Foley 1976), particularly in instances of nominal modification. Scontras and Nicolae (2014) propose that the *-ng* linker of Tagalog nominal modification effectively performs Predicate Modification with its inputs, simplifying the rules of semantic composition to only Functional Application:

(62) Tagalog:  $\llbracket \text{LK} \rrbracket = \lambda P_{\langle e,t \rangle}.\lambda Q_{\langle e,t \rangle}.\lambda x.P(x) \wedge Q(x) = 1$  (Scontras and Nicolae 2014, 23)

The distribution of the Tagalog linker is vastly different from that of the Javanese linker. The Tagalog linker is present in a bevy of nominal modification constructions: attributive adjectives, relative clauses, adverbial modifiers, and nominal modifiers. On the other hand, in Javanese the linker only attaches to numerals. Thus, the proposal for the Javanese linker is not incompatible with the Tagalog proposal. Rather, it is a caution against importing the analysis of one language's linker into that of another language, even when the linker is homophonous and the languages are in the same typological subgroup.

Notably, however, Scontras and Nicolae (2014) leave open the possibility that the *-ng* linker appearing on Tagalog numerals is distinct from the homophonous linker appearing in other modification constructions in the language. As in Javanese, linker-marked numerals in Tagalog must precede the noun being modified.

(63) isa-ng babae  
 one-LK woman  
 'one woman' Tagalog

(Scontras and Nicolae 2014, 35a)

I leave for future work a direct comparison between the linker *-ng* that appears on Javanese numerals and the linker *-ng* that appears on Tagalog numerals.

## 4.2. Two potential paths for simple numerals

To round out the structure and composition of Javanese numerals, I consider two analyses of simple numerals: that they have basic number denotations, and that they have basic property denotations. Though the former analysis enjoys fewer stipulations than the latter, neither faces immediate empirical challenges.

### 4.2.1. A basic- $n$ denotation of simple numerals

Suppose that simple numerals like *telu* ‘three’ had basic denotations of abstract numbers. Simple-numeral arguments to unit nouns readily undergo Functional Application. For simple-numeral modifiers, however, there must be a means of deriving the  $\langle e, t \rangle$  denotation for such numerals in cases as in (64).

- (64) Ana bocah lanang cilik~cilik telu  
 exist child male small~RED three  
 ‘There are three small boys.’

(65)  $\llbracket \text{bocah lanang cilik~cilik telu} \rrbracket = \lambda x. \text{CHILD}(x) \wedge \text{MALE}(x) \wedge \text{SMALL}(x) \wedge |x| = 3$

If there were a syntactically free rule of semantic composition to derive a type- $\langle e, t \rangle$  denotation from a basic- $n$  denotation, this rule would overgenerate. Namely, it would wrongly predict simple-numeral modifiers to linearly precede the head noun. To syntactically constrain simple-numeral modifiers to post-nominal positions, I recall that the semantic type of simple numerals corresponds with their word-order restrictions. Type- $\langle e, t \rangle$  numerals must follow the head noun, while type- $n$  numerals must precede the measurement unit. I consider a covert modifier MANY as in (66) to shuttle basic- $n$  simple numerals to derived- $\langle e, t \rangle$  denotations.

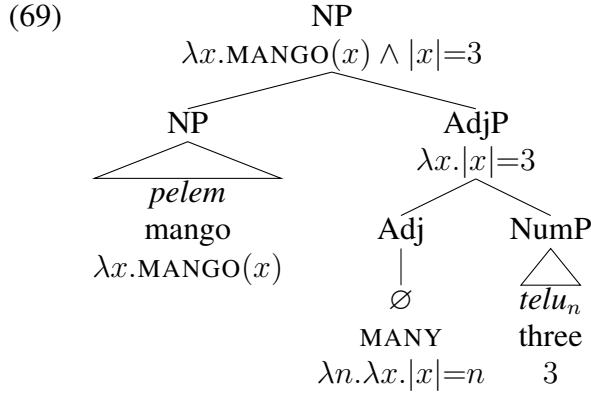
(66)  $\llbracket \text{MANY} \rrbracket = \lambda n. \lambda x. |x| = n$  (Buccola’s (2015) ISCARD, 47)

Supplied with a number-denoting *telu<sub>n</sub>* ‘three,’ the complex adjective denotes the property of having cardinality three.

(67)  $\llbracket \text{telu}_n \rrbracket = 3$

(68)  $\llbracket \text{MANY} \rrbracket (\llbracket \text{telu}_n \rrbracket) = \lambda x. |x| = 3 = \llbracket \text{telu}_{\langle e, t \rangle} \rrbracket$

This MANY is obligatorily post-nominal, for the same reason that all adjectives and demonstratives in Javanese are post-nominal (Vander Klok 2009). While simple numerals must compose with post-nominal MANY in order to be modifiers, complex numerals do not—they are built to be property-denoting. A covert MANY not only has explanatory power for why simple-numeral modifiers must be post-nominal; it also correctly predicts that complex-numeral modifiers need not be post-nominal. Explicitly, simple-numeral modifiers are supplied to a covert MANY operator to form an adjunct to the extended nominal projection as in (69). Post-nominal simple numerals are correctly predicted to not bear the linker, while pre-unit simple numerals are.



Taking the  $\text{CARD}_{\text{NEW}}$  operator to be syntactically present between the complex-numeral argument and the unit noun, the preceding syntactic account of the linker can correctly predict the lack of linker-marking on complex numerals. The basic- $n$  approach to simple numerals is compatible with two different explanations for the lack of linkers: (i) linker-marking arises from a strictly local Agree relation between the unit noun and the argument, and  $\text{CARD}_{\text{NEW}}$  intervenes as shown in (52); or (ii) linker-marking only arises when an Agree relation holds between the unit noun and an NumP, and complex numerals are not themselves NumPs.

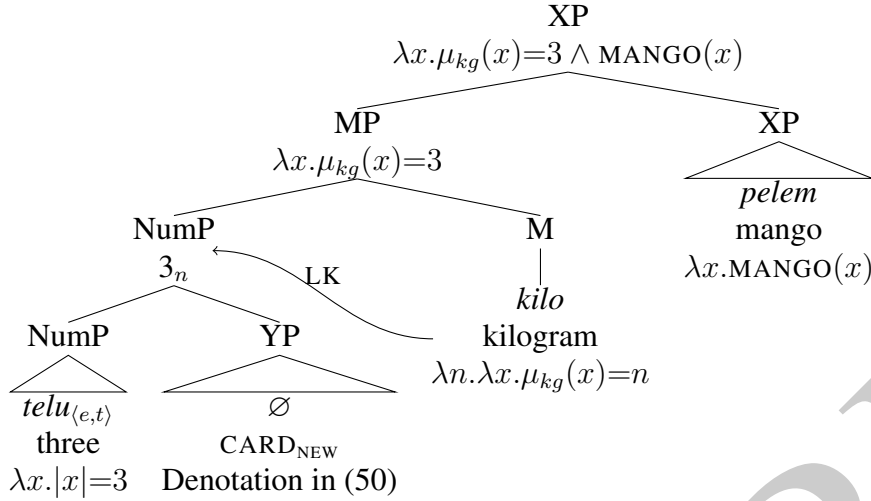
Under this basic- $n$  approach to simple numerals, the  $\text{CARD}_{\text{NEW}}$  operator is only needed for complex numerals, while the  $\text{MANY}$  operator is only needed for simple numerals. I consider the consequences of having two operators at work, but first I lay out an alternative approach that opts for a single operator.

#### 4.2.2. A basic- $\langle e, t \rangle$ denotation of simple numerals

Suppose instead that simple numerals, like complex numerals, have basic property denotations. Then the case of simple-numeral modifiers requires no  $\text{MANY}$  operator. Simple numerals are themselves beholden to the same restriction on adjectives and demonstrative modifiers in the language, with simple numerals' syntactic category explaining the word-order restriction on simple-numeral modifiers. Instead, there must be a means of shuttling type- $\langle e, t \rangle$  numerals to type- $n$  denotations. With  $\text{CARD}_{\text{NEW}}$  already motivated for complex-numeral arguments, the same operator can be employed for simple numerals. By this approach as in (70), any input to a unit noun in Javanese will itself be the output of composition with a  $\text{CARD}_{\text{NEW}}$  operator. This analysis of numeral type is committed to a particular analysis of the linker: linkers cannot be a reflex of strict adjacency between the unit noun and the numeral. The strict-adjacency account struggles because  $\text{CARD}_{\text{NEW}}$  intervenes between the numeral and the unit noun, in the case of both simple and complex numerals. Rather, the linker's distribution must appeal to a syntactic-categorical distinction between simple numerals and complex numerals: the linker attaches to NumPs (i.e., simple numerals), but not MPs (complex numerals).



(70)



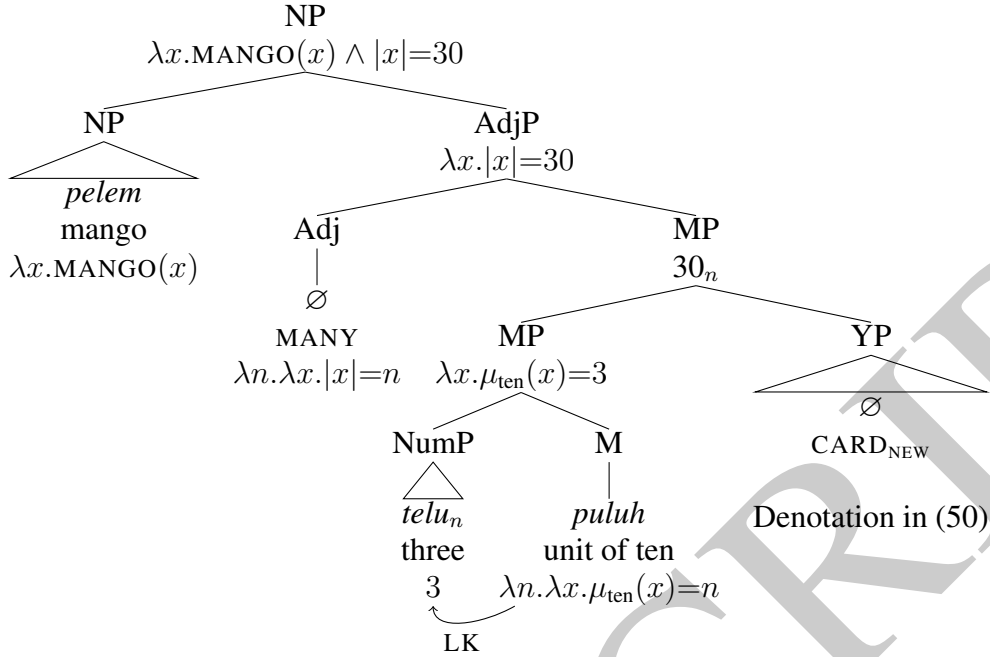
### 4.3. Comparing simple-numeral accounts: the case of post-nominal complex-numeral modifiers

The preceding two analyses of simple numerals' basic type place different demands on syntax and semantics of Javanese nominals. The first, a basic- $n$  analysis, requires two operators: *MANY* for simple numerals and its inverse *CARD<sub>NEW</sub>* for complex numerals. It does not require a distinction in numerals' syntactic categories to explain the distribution of the linker; the distribution and intervention of *CARD<sub>NEW</sub>* could alone explain the systematic lack of linkers on complex-numeral arguments within MPs. By this first analysis, numerals in Javanese show internal heterogeneity not only in their morphosyntactic buildup but also in their basic semantic type.

The second analytical option, a basic- $\langle e, t \rangle$  analysis, requires just one operator: *CARD<sub>NEW</sub>*. However, it requires a distinction in numerals' syntactic categories to explain the distribution of the linker; *CARD<sub>NEW</sub>* always intervenes between numeral arguments and unit nouns. By this analysis, numerals in the language have heterogeneous morphosyntax but show homogeneity in their basic semantic type.

Adjudicating between these two options is independent of the main theoretical claim in this paper: numerals cannot unilaterally be given a basic type- $n$  denotation. Empirically, neither of the two preceding analyses of simple numerals has been falsified. A natural line of reasoning would be to revisit a concern raised with the basic- $n$  account: the consequences of having *MANY* and its inverse, *CARD<sub>NEW</sub>*, both operative. Is it problematic that *MANY* and *CARD<sub>NEW</sub>* could work together, where the type- $n$  output of *CARD<sub>NEW</sub>* is itself an input to *MANY*? I call this *STACKING*, represented as in (71).

(71) *STACKING* to obtain a post-nominal complex-numeral modifier

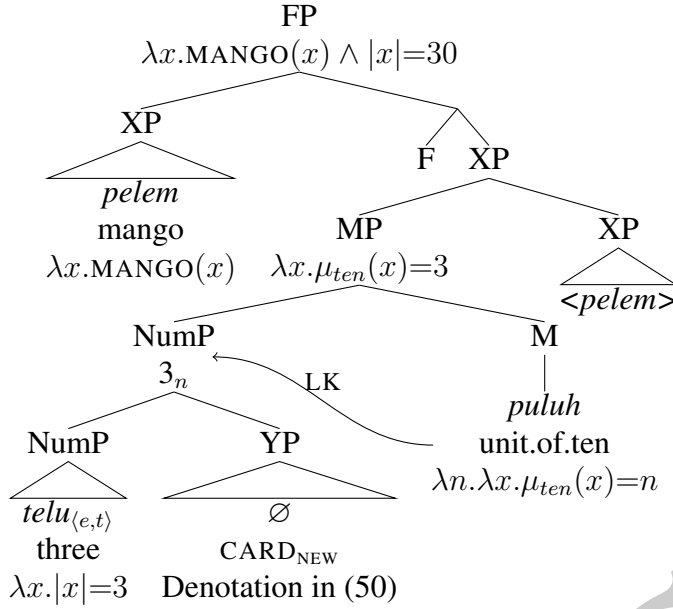


The concern raised by STACKING is one of overgeneration, but is it genuinely overgeneration? Recall that the linearized output of a structure like (71) is well-formed: complex-numeral modifiers can linearly follow the head noun. However, such post-nominal complex-numeral modifiers could be derived by other means. Namely, the flexible constituent order with complex-numeral modifiers could be accounted for by STRANDING. Stranding is observed for other MPs, those that are not complex numerals. By the stranding approach, the nominal projection to which the MP adjoins is moved higher in the nominal or clausal projection. The MP is left behind. Certain passive-like constructions in Javanese can be treated as more dramatic cases of clause-level stranding: in (72), the demonstrative-marked noun phrase *beras-e* ‘the uncooked rice,’ is moved to the clause-initial position. Meanwhile, the MP *pat-ang karung* ‘four sackfuls’ remains in the post-verbal position.

- (72) Beras-e            wis        ta'-tuku        pat-ang karung  
 uncooked.rice-DEF already 1SG.NAV-buy four-LK sack  
 ‘The uncooked rice, I already bought four sackfuls.’

The observed flexibility of MP position—particularly for post-nominal complex-numeral modifiers—could be treated as more modest cases of MP-stranding, assuming that there is a low landing site, such as a DP-internal focus position in the specifier of FP, to which the nominal projection moves. The stranding approach could be integrated into the basic- $n$  analysis or into the basic- $\langle e, t \rangle$  analysis.

- (73) STRANDING to obtain a post-nominal complex-numeral modifier



Returning to the question of simple numerals' basic type in Javanese, the choice between a basic- $n$  versus basic- $\langle e, t \rangle$  analysis can be reframed as the choice between whether STACKING is empirically necessary: the affirmative would support the basic- $n$  analysis since the basic- $\langle e, t \rangle$  account gives no means to smuggle a complex numeral into the argument of MANY. Put another way, STACKING predicts that complex numerals can appear anywhere that simple numerals can: the distribution of complex numerals is a superset of the distribution of simple numerals.

Testing against STACKING calls for syntactic signposts that must attach higher than MANY but lower than MPs. Inventorying the high reaches of DPs—demonstratives, definites, and possessives; the former exemplified in (74-75)—there is not an attested syntactic environment in which simple numerals are allowed but complex numerals are not. While a null result for a testing between STACKING and STRANDING, this result may itself be informative for Javanese DP syntax. The data are consistent with simple- and complex-numeral modifiers attaching at the same syntactic height but with asymmetric syntactic consequences: simple-numeral modifiers obligatorily attach to the right, while complex-numeral modifiers attach to the left and can be stranded. The stranding asymmetry is consistent with general syntactic restrictions on displacement: the element displaced in DP-internal displacement must include the head noun ("Cinque's constraint" as noted in [Ouwayda and Shlonsky 2017](#)), and displacement is a leftward operation. These findings are compatible with either analysis of simple numerals' basic type. Independent of this choice, the Javanese data pose a challenge for unilaterally treating all numerals with basic type- $n$  denotations.

- (74) *Adi nge-lap-i pelem { telu / telu-ng puluh } kuwe*  
 Adi AV-clean-APPL mango three three-LK unit.of.ten DEM  
 'Adi cleans those {three / thirty} mangos.'

- (75) \**Adi nge-lap-i pelem kuwe { telu / telu-ng puluh }*  
 Adi AV-clean-APPL mango DEM three three-LK unit.of.ten  
 Intended: 'Adi cleans those {three / thirty} mangos.'<sup>13</sup>

<sup>13</sup>For both simple and complex numerals in (75), the quality of the sentence is improved by a marked pause between the demonstrative and the numeral.

There is one logically possible analysis that is not yet mentioned: simple numerals having basic- $\langle e, t \rangle$  denotations and complex numerals having basic- $n$  denotations. Such an analysis would involve more fundamental changes to the system built so far, and I sketch the analysis before providing a general discussion.

#### 4.4. Basic- $\langle e, t \rangle$ simple numerals, basic- $n$ complex numerals

Here, I suppose that simple numerals have basic- $\langle e, t \rangle$  denotations and complex numerals have basic- $n$  denotations. In this analysis, counting bases like *puluh* ‘unit of ten’ must not have type  $\langle n, \langle e, t \rangle \rangle$  as has been assumed earlier; otherwise, *puluh* will take a number and return a complex numeral as an  $\langle e, t \rangle$  property, contradicting the initial assumption. Suppose, then, that *puluh* is a function from numbers to numbers. Then the semantics of the linker becomes immediate: the linker is the realization of  $\text{CARD}_{\text{NEW}}$ : it attaches to type- $\langle e, t \rangle$  simple numerals to shuttle them to type- $n$  denotations, and it does not attach to complex-numeral arguments (33) because they are already of type  $n$ .

What remains is to account for modifier usage of complex numerals: there must be either a compositional rule or an operator like *MANY* to combine with type- $n$  complex numerals. A syntactically free compositional rule would over-generate: by permitting type- $n$  complex numerals to combine post-nominally with lexical nouns, type- $n$  linker-marked simple numerals are wrongly predicted to be well-formed as post-nominal modifiers. A syntactically constrained  $\langle n, \langle e, t \rangle \rangle$  operator like *MANY* would need to select only type- $n$  complex numerals and not type- $n$  linker-marked simple numerals, to avoid over-generation. That is, at some level of representation, complex numerals must have distinct representations from linker-marked simple numerals. This distinction is crucial to capture the morphosyntactic asymmetries in numerals’ modifier usage: the syntactic mechanism allowing pre-nominal and post-nominal complex-numeral modifiers must be barred from allowing the same configurations for simple-numeral modifiers. The distinction could be in syntactic category, or—if we entertain a different starting point—semantic type: that complex numerals basic type is neither  $\langle e, t \rangle$  nor  $n$ .

Multiplicative complex numerals, and the counting bases that output them, evoke questions about the ontological relationship between numbers and degrees: whether, for instance, degrees act as a mediator between numbers and individuals (Lønning 1987; Champollion 2017, Ch. 2.4-2.5 and references therein). Degrees are not formally represented in the present paper’s semantic derivations. However, complex numerals appear to offer a natural entryway for studying degrees vis-à-vis numbers in Javanese, as their morphology and syntax shows similarities with both categories. Complex numerals like *telu-ng puluh* ‘thirty’ appear in places expected of numbers (e.g., the argument of *kilo* ‘kilogram’) while themselves having the morphological buildup of phrases analyzable as degrees (e.g., *telu-ng kilo* ‘three kilograms’). How Javanese numerals bear on the ontological status of degrees, I identify for upcoming work.

#### 4.5. General discussion

In this section, I have presented two means of composing Javanese cardinal numerals under the assumption that the available basic types for numerals are either  $n$  or  $\langle e, t \rangle$ . In both of these analyses as well as the final consideration of basic  $\langle e, t \rangle$  simple numerals, basic- $n$  complex numerals in §4.4, two generalizations remain: (i) while there are several tenable analytical starting points for numerals’ basic types in Javanese, it is empirically problematic to treat all numerals as having

a basic type- $n$  denotation; (ii) the  $\text{CARD}_{\text{NEW}}$  operator is present in each analysis. While [Bylinina and Nouwen \(2020\)](#) present this  $\langle e, t \rangle$ -to- $n$  pathway as an analytical option, they remark that the approach has not been widely used in numeral semantics cross-linguistically. Javanese presents evidence that such a pathway is indeed useful in numeral semantics.

## 5. Conclusion

I have motivated the constituency and composition of complex numerals in Javanese DPs. Along the way, I mapped numerals' linear position to their structural position and accompanying semantic role. I also identified the syntactic sensitivities of the linker suffix. I identify two implications: one for Javanese syntax and another for cross-linguistic numeral theory.

### 5.1. Constituency in pseudo-partitives and partitives

The MP itself is never shown to dislocate from its base-generated position, unless the entire nominal projection containing it is fronted, as in (76).

- (76) [[Pat-ang karung] beras]            wis    ta'-tuku  
 four-LK sack    uncooked.rice already 1SG.NAV-buy  
 'I already bought four sackfuls of uncooked rice.'

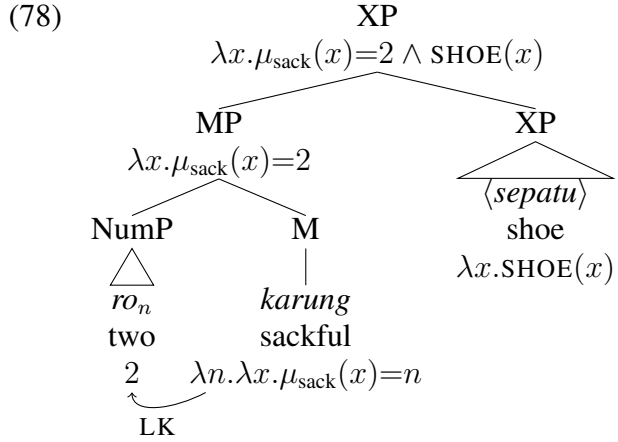
Notably, the noun being measured over can be absent entirely in contexts where it is sufficiently clear from context.

- (77) Ro-ng    karung di-pikul                    mbiyen wong    langka    becak  
 two-LK sack    NAV-carry.on.the.shoulders past    person not.exist pedicab  
 'Two sacks [of the shoes] were carried on shoulder as there was no pedicab.'  
 (PEM-130402.0761)<sup>14</sup>

The canonical cases of measurement in this paper, as in *telu-ng kilo pelem* 'three kilograms of mangos,' display three identifiable ingredients: a number-denoting argument like *telu* 'three,' a unit noun like *kilo* 'kilogram,' and a substance noun like *pelem* 'mango.' The sentence in (77) is interpreted as having all three parts, even though the substance noun *sepatu* 'shoes' is not pronounced in this utterance; it was uttered in a preceding sentence. Thus, (77) is still compatible with the structure of MPs motivated in this paper, shown in (78).<sup>15</sup>

<sup>14</sup>The linker-marked form *ro-ng* for the numeral 'two' might seem surprising given that the linker-less form is *loro*. Historically, the form *loro* descends from *roro*, a reduplication of *ro* ([Widada et al. 2001](#)). For this reason, *rong* can be analyzed as *ro-ng*. I make the non-crucial choice to represent the post-nominal form *loro* without reduplication in the glosses. A very similar situation is observed for 'four': the linker-marked form is *pat-ang* while the modern linker-less form is *papat*.

<sup>15</sup>For simplicity of presentation, I assume *karung* to denote 'sackful,' abstracting away how this measurement-unit construal is obtained from the homophonous ordinary noun meaning 'sack.' See §2.1 for a discussion of a parallel pattern with *kranjang* 'basket.'



It is unclear, however, what level of structure the implicit substance is projecting. In the context of the utterance, the implied shoes are ‘those shoes,’ the ones being talked about. The SUBSTANCE should then be a syntactic unit large enough to contain a demonstrative or another contextually determined anaphoric variable. The ability for MPs to combine with bare nouns as well as demonstrative-marked nouns invites further research into the structure of Javanese pseudo-partitives and partitives.

Descriptively and analytically, sentences like (76) and (77) are striking because they are thematic undergoers that occupy the “surface-subject” position of the clause, preceding agents and auxiliaries; yet they are not overtly marked for “definiteness/specificity.” More concretely, they do not have overt demonstratives or other overt markers associated with definiteness. The label “definiteness/specificity” is used in Austronesian syntax (e.g., [Patrianto and Chen 2023](#) for Javanese in particular) to characterize some languages’ restrictions on the undergoers that can occupy the surface-subject syntactic position. To my knowledge, in Javanese the only reported cases of undergoer surface-subjects that lack demonstratives and that are not themselves proper names or implicitly possessed are undergoers with DP-initial, linker-marked numerals. MPs’ high attachment, adjoining to the DP, could help clarify why undergoers with pre-unit numerals can appear in the surface-subject position. In turn, the “definiteness/specificity” label may come to be made more precise: for instance, whether the requirement solely relates to the undergoer having a particular syntactic height, and what that height is.

## 5.2. Constituency in and of numerals

In Javanese, complex numerals are constituents that are relatively transparent in their morphological composition: *telu-ng puluh* ‘thirty’ is literally ‘three tenfuls.’ In contrast, English *thirty* faces a less clear path—if any—to a compositional analysis in the synchronic grammar. [Ionin and Matushansky \(2018\)](#) report that complex numerals are not constituents in themselves. They draw on data from over a dozen languages, including Dagaare (Niger-Congo), Russian (Indo-European), and Basque (isolate). The claims of the present paper—namely, that Javanese complex numerals are constituents—are not incompatible with [Ionin and Matushansky’s \(2018\)](#), or with [Tatsumi’s \(2025\)](#) syntactic generalizations. Russian, Javanese, and English could be taken as case studies in different grammaticalizations of numerals. While [Ionin and Matushansky \(2018\)](#) offer evidence against Russian complex numerals as constituents, this paper shows that for Javanese the constituency does hold. For English, the question might not be whether complex numerals are constituents, but rather *whether complex numerals are part of the narrow linguistic grammar at*

*all*; Bylinina and Nouwen (2020) and Morzycki (2017) consider how technical terms like certain numerals or other mathematical objects appear to abstain from semantic composition. In studying numeral semantics and language variation, the Javanese constituency data should not be held to bear directly on the constituency of numerals cross-linguistically. Rather, Javanese offers a lesson on what cannot hold of a universal theory of numerals. Namely, numeral theory must allow for complex numerals to be constituents, and it cannot unilaterally require a type- $n$  denotation as basic for all numerals.



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